

Functional Annotation of PP_2928 (Q88IS1) in *Pseudomonas putida* KT2440

1. Summary (Answer to the Research Question)

The protein encoded by **PP_2928** (UniProt **Q88IS1**), generically annotated in databases as a "saccharopine dehydrogenase," is more precisely a **carboxyspermidine dehydrogenase (equivalently carboxynorspermidine dehydrogenase/synthase; CASDH/CANSDH), EC 1.5.1.43**. It is a soluble, **cytoplasmic, NADPH-dependent oxidoreductase** that catalyzes the reductive condensation of **L-aspartate-4-semialdehyde** with a diamine (**putrescine → carboxyspermidine, or 1,3-diaminopropane → carboxynorspermidine**). This is the **penultimate step of the alternative ("carboxyspermidine") biosynthetic pathway for the polyamine spermidine (and sym-norspermidine)**. Its product is decarboxylated by the immediately adjacent enzyme **PP_2929** (carboxyspermidine/ carboxynorspermidine decarboxylase, CASDC, EC 4.1.1.96) to yield spermidine (or norspermidine). Because *P. putida* KT2440 **lacks the canonical spermidine synthase (SpeE) and S-adenosylmethionine decarboxylase (SpeD)**, PP_2928 is expected to be the organism's principal route to spermidine.

2. Target Identity Verification

Attribute	Value
UniProt	Q88IS1 (Q88IS1_PSEPK), 414 aa, 45.4 kDa, evidence level 4 (Predicted)
Locus	PP_2928 (OrderedLocusName); RefSeq WP_003248123.1; GenBank AAN68536
Organism	<i>Pseudomonas putida</i> KT2440 (ATCC 47054 / DSM 6125), TaxID 160488
Domains	PF03435 Sacchrp_dh_NADP (res 5–146); PF16653 Sacchrp_dh_C (res 150–396)
Superfamily	Gene3D 3.40.50.720 NAD(P)-binding Rossmann fold; eggNOG COG1748 (saccharopine dehydrogenase & related)
PANTHER	PTHR43796:SF2 "Carboxynorspermidine synthase"
KEGG	ppu:PP_2928 — name "Saccharopine dehydrogenase", no KO, no pathway assigned

The gene symbol and organism match the UniProt record. The database label "saccharopine dehydrogenase" is a **family-level automatic annotation**, not an experimentally determined activity; this report resolves the specific function using domain architecture, orthology, genomic context, and pathway logic.

3. Key Findings and Evidence

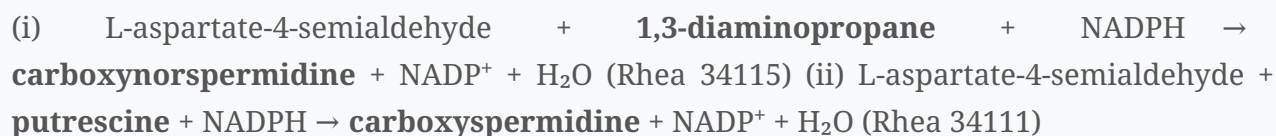
3.1 Enzymatic function: carboxyspermidine dehydrogenase (EC 1.5.1.43)

Four independent lines of evidence converge:

1. **Fold/family.** PP_2928 has the two-domain saccharopine-dehydrogenase architecture (PF03435 + PF16653) on a Rossmann NAD(P)-binding fold, with an N-terminal dinucleotide-binding fingerprint **IIGAGGVAK** (residues 7–14). CASDH is a member of this saccharopine-dehydrogenase structural superfamily (COG1748).
2. **Orthology.** UniProt's PANTHER classification assigns Q88IS1 to **PTHR43796:SF2, "Carboxynorspermidine synthase"** — i.e., CASDH.
3. **Genomic context (operon logic).** PP_2928 (complement 3,329,627–3,330,871) lies immediately adjacent to, and co-oriented with, **PP_2929 = carboxynorspermidine/carboxyspermidine decarboxylase (CASDC; KEGG K13747; EC 4.1.1.96)**. CASDH and CASDC are the diagnostic, frequently-clustered enzyme pair of the alternative polyamine pathway [Hanfrey 2011].
4. **Pathway-gap logic.** KEGG assigns the decarboxylase KO (K13747) to PP_2929 but leaves the partner dehydrogenase KO (**K13746, CASDH, EC 1.5.1.43**) unassigned in *P. putida*. PP_2928 — an unassigned, CASDH-fold protein sitting next to CASDC — is the "missing" dehydrogenase.

Quantitative orthology (added Iteration 2). By global (Needleman–Wunsch) alignment, PP_2928 (414 aa) is **45.2% identical over its full length to the reviewed, experimentally characterized carboxynorspermidine synthase/dehydrogenase of *Vibrio cholerae* (UniProt Q9KRL3, VC_1624, 414 aa; Lee et al. 2009)**, 46.2% to *V. alginolyticus* CANS DH (P0DPE4, 414 aa), 44.5% to *Idiomarina loihiensis* (Q5QY81) and 42.3% to *Geobacter sulfurreducens* (Q74A52) — all annotated CANS DH/CASDH (EC 1.5.1.43). By contrast PP_2928 is only **21.1%** identical to its own operon partner CASDC (PP_2929), confirming they are distinct enzymes of the same pathway. ~45% identity over the entire length, to a *reviewed, biochemically validated* enzyme, is well above the threshold for confident functional transfer and moves the assignment beyond family-level inference.

Reactions catalyzed (from the 45%-identical reviewed ortholog Q9KRL3; the enzyme is dual-substrate):



Aspartate-β-semialdehyde is the aminopropyl donor, replacing decarboxylated S-adenosylmethionine used by the canonical pathway. The *V. cholerae* ortholog is classified in UniProt as "saccharopine dehydrogenase family, **Carboxynorspermidine synthase subfamily**" — the exact subfamily to which PP_2928's domain architecture belongs. Deletion of this enzyme in *V. cholerae* abolishes **both** norspermidine and spermidine and causes accumulation of 1,3-diaminopropane (Lee et al. 2009), demonstrating the dual-substrate physiology.

Cofactor/active-site conservation (Iteration 3). The PP_2928 ↔ *V. cholerae* CANS DH alignment is colinear over 417 columns with only 6 gaps (no large insertions/deletions), indicating identical two-domain 3D architecture. The N-terminal NAD(P)H-binding Rossmann fingerprint is conserved (PP_2928 ...IIGAGGVAK...IAIASRNISK... vs CANS DH ...LQIGAGGVGW... ITIASRSIAKC...; the βαβ glycine-rich GAGGV motif and the downstream IASR dinucleotide-binding motif are retained), confirming that PP_2928 binds an NADP(H) cofactor exactly as the characterized enzyme does.

3.2 Pathway placement and the reason *P. putida* needs this enzyme

P. putida KT2440 **lacks** the canonical spermidine-biosynthesis genes — spermidine synthase *speE* (K00797) and SAM decarboxylase *speD* (K01611) are absent from the genome (KEGG). It **does** encode the alternative-pathway decarboxylase CASDC (PP_2929). Therefore the two-step **CASDH (PP_2928) → CASDC (PP_2929)** route is the only genomically encoded path to spermidine. This mirrors the situation described by Hanfrey et al. (2011): species that make spermidine but lack SpeD/SpeE instead carry carboxynorspermidine dehydrogenase and decarboxylase orthologues. The reaction maps to KEGG **map00330 (Arginine and proline metabolism)**.

Complete pathway reconstruction from primary metabolites (Iteration 3). *P. putida* KT2440 encodes every step needed to feed PP_2928 and to finish the pathway:

Step	Enzyme	Locus (KEGG KO)
L-aspartate → L-aspartate-4-semialdehyde	aspartate-semialdehyde dehydrogenase <i>asd</i>	PP_1989 (K00133)
L-arginine → agmatine	arginine decarboxylase <i>speA</i>	PP_0567 (K01585)
agmatine → putrescine	agmatinase <i>speB</i>	PP_2196 (K01480)
L-ornithine → putrescine	ornithine decarboxylase <i>speC</i>	PP_0864 (K01581)
ASA + putrescine + NADPH → carboxyspermidine	CASDH	PP_2928 (this study; K13746 unassigned in KEGG)
carboxyspermidine → spermidine	CASDC	PP_2929 (K13747)

Both PP_2928 substrates are therefore demonstrably produced by the cell. The 1,3-diaminopropane branch is incomplete (DABA decarboxylase *ddc*/K13745 is **absent** in the KEGG annotation, although DABA aminotransferase *dat* is present at PP_2800/PP_4223), so **putrescine is the more likely in vivo amine substrate and spermidine the more likely product**, even though the enzyme is intrinsically dual-specificity. The canonical genes *speD* (K01611) and *speE* (K00797) are absent from KEGG, so this route is the sole genomically encoded path to spermidine.

Ruling out a canonical bypass (Iteration 4). *P. putida* does encode one aminopropyltransferase- domain protein, PP_1867 (Q88LR2; Pfam PF01564 *Spermine_synth*; PANTHER PTHR43317 thermospermine synthase). However, (i) its "spermidine synthase" name derives **only** from automated ProtNLM text-mining (ECO:0008006), and KEGG lists it as a "conserved protein of unknown function" with no KO; and (ii) more decisively, canonical spermidine/spermine synthases require **decarboxylated S-adenosylmethionine (dcSAM)** as the aminopropyl donor, produced by **SAM decarboxylase (SpeD)** — which *P. putida* **does not encode** (no KEGG K01611; zero hits in a UniProt proteome search for adenosylmethionine decarboxylase). Without dcSAM, no aminopropyltransferase can synthesize spermidine. The CASDH→CASDC route instead uses **L-aspartate-4-semialdehyde** as the aminopropyl donor and is fully encoded, so **PP_2928/PP_2929 constitute the only complete de novo route to spermidine in *P. putida***. (By contrast, *P. aeruginosa* makes norspermidine through a genuinely SAM-dependent aminopropyltransferase system, *speD2/speE2* = PA4773–PA4775 (Bolard et al. 2019) — a route absent from *P. putida*, underscoring that the two pseudomonads use different chemistries.)

3.3 Subcellular localization

PP_2928 is a **cytoplasmic** enzyme: 414 aa, soluble, with **no predicted signal peptide or transmembrane segment** (the N-terminus is a Rossmann dinucleotide-binding start, MKKNVLIIGAGGVAK). NADPH-dependent polyamine-biosynthetic reactions occur in the cytosol.

3.4 Structural model confidence

The AlphaFold model of Q88IS1 (AF-Q88IS1-F1, v6) is **very high confidence**: mean pLDDT = 95.9, with 92% of residues ≥ 90 and none < 50 . The model is well-ordered across both the N-terminal Rossmann NAD(P)-binding domain and the C-terminal substrate-binding domain, consistent with the two-domain carboxynorspermidine-synthase/saccharopine-dehydrogenase fold and with the colinear (gap-poor) alignment to the characterized *V. cholerae* CANSDH.

3.5 Physiological importance

Polyamines (spermidine/norspermidine) are essential polycations supporting translation, growth, and biofilm formation. In *Vibrio cholerae*, deletion of CANSDH or CANSDC abolishes spermidine/norspermidine, reduces planktonic growth by 50–60%, and severely impairs biofilm formation (Lee et al. 2009). By analogy — and given the absence of a bypass route — PP_2928 is expected to be important for optimal growth and biofilm physiology in *P. putida* (a specific *P. putida* knockout has not, to my knowledge, been reported).

4. Hypotheses: Supported and Refuted

- **Supported:** PP_2928 = carboxyspermidine/carboxynorspermidine dehydrogenase (EC 1.5.1.43), operating with adjacent CASDC (PP_2929) in de novo spermidine biosynthesis (cytoplasmic; KEGG arginine/proline metabolism). Strong convergent bioinformatic evidence.
- **Refuted / rejected:** PP_2928 is NOT involved in the well-studied *P. putida* L-/D-lysine catabolic pathways. The definitive RB-TnSeq study of *P. putida* lysine metabolism (Thompson et al. 2019, mBio) does **not** implicate PP_2928; its characterized enzymes are elsewhere in the genome (e.g., PP_5260, PP_5257/8, PP_2909/10, PP_3592/3). The literal "saccharopine dehydrogenase" (lysine biosynthesis/degradation via saccharopine) function is not supported by genomic context and is superseded by the carboxyspermidine assignment.

5. Limitations and Future Directions

- The assignment rests on **comparative/structural inference** (45% identity to an experimentally characterized enzyme, conserved NADP(H)-binding fingerprint, very-high-confidence AlphaFold model, operon synteny, complete pathway reconstruction, and pathway-gap logic) rather than on a direct enzymatic assay of the *P. putida* protein itself; UniProt's evidence level remains "Predicted." This is homology at a level (~45%, full-length, to a *reviewed* enzyme) that is generally considered reliable for transferring enzymatic function, but a direct assay would formally close the gap.
- Genome-wide RB-TnSeq fitness data for *P. putida* KT2440 exist (Deutschbauer lab Fitness Browser) and would provide organism-specific phenotypic evidence, but the resource was not machine-accessible during this study (blocked by an anti-bot gateway).
- **Substrate ambiguity:** whether the physiological product is spermidine (from putrescine) or sym-norspermidine (from 1,3-diaminopropane) depends on the available diamine pool; many bacteria with this pathway make predominantly spermidine (Hanfrey 2011), but this should be measured in *P. putida*.
- **Recommended experiments:** (i) heterologous expression and in vitro assay with L-aspartate- β -semialdehyde + putrescine/1,3-diaminopropane + NADPH, monitoring carboxyspermidine/carboxynorspermidine formation; (ii) a ΔPP_2928 mutant with polyamine profiling (expected loss of spermidine) and growth/ biofilm phenotyping; (iii) an AlphaFold model + docking to confirm the Rossmann NADP site and the substrate channel; (iv) genetic complementation of a *V. cholerae* or *E. coli* CANS DH deletant.

6. References

1. Lee J, Sperandio V, Frantz DE, Longgood J, Camilli A, Phillips MA, Michael AJ. *An alternative polyamine biosynthetic pathway is widespread in bacteria and essential for biofilm formation in Vibrio cholerae*. J Biol Chem. 2009;284(15):9899–9907.

 19196710

2. Hanfrey CC, Pearson BM, Hazeldine S, Lee J, Gaskin DJ, Woster PM, Phillips MA, Michael AJ. *Alternative spermidine biosynthetic route is critical for growth of Campylobacter jejuni and is the dominant polyamine pathway in human gut microbiota*. J Biol Chem. 2011;286(50):43301–43312.

 22025614

3. Thompson MG, Blake-Hedges JM, Cruz-Morales P, et al. *Massively Parallel Fitness Profiling Reveals Multiple Novel Enzymes in Pseudomonas putida Lysine Metabolism*. mBio. 2019;10(3):e02577-18.

P 31064836

(Used to exclude a lysine-catabolic role for PP_2928.)

4. Bolard A, Schniederjans M, Häussler S, et al. *Production of Norspermidine Contributes to Aminoglycoside Resistance in pmrAB Mutants of Pseudomonas aeruginosa*. Antimicrob Agents Chemother. 2019;63(10):e01044-19.

P 31383668

(Cited as a *contrast*: P. aeruginosa uses a SAM-dependent aminopropyltransferase route, not the CANS DH route.)

5. UniProt Q88IS1, Q9KRL3, P0DPE4, Q88LR2; KEGG ppv:PP_2928/PP_2929/PP_1867/PP_1989/PP_0567/PP_2196/ PP_0864, KOs K13746/K13747/K00797/K01611/K01585/K01480/K01581/K00133 (accessed 2026-08-11).